Chapter 11

Variational Formulation of Elliptic Problems

We begin our investigation and our initial formulation of finite element methods with a bias toward elliptic problems. The reason is that we can often relate uniform ellipticity of a PDE to the well-posedness of the corresponding boundary-value problem in a relatively straightforward way. In later sections, we extend our analysis to more general cases. It is recommended that you review basic facts about Sobolev spaces before continuing.

11.1 Motivation: The Poisson Problem

First, we return to the general elliptic problem already considered in Chapter 10, which we now formally define as

Model Problem 4 Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with a smooth boundary, and let $A = (a_{ij})_{i,j=1,\dots,d}$, where $a_{ij} \in L^\infty(\Omega)$ are bounded functions, be uniformly elliptic in the sense of Definition 10.1. Let $f \in L^2(\Omega)$ and consider the homogeneous boundary value problem

$$\begin{aligned} -\nabla \cdot (A(x)\nabla u) &= f, \qquad && x \in \Omega, \\ u &= 0, && x \in \partial\Omega. \end{aligned} \tag{11.1}$$

(Note that we require that the a_{ij} are merely bounded functions, not necessarily differentiable. This will be justified by considering the weak formulation, shifting the differentiation to test functions.)

Similar to our approach for hyperbolic PDE, we derive and analyze the weak formulation. It turns out that this will lead us to strong results, under modest assumptions regarding the regularity of the data and the solution.

First, choose an arbitrary $v \in C_0^\infty$, and integrate (11.1) by parts:

$$-\int_\Omega \nabla \cdot (A(x)\nabla u) v \, dx = \int_\Omega (A(x)\nabla u) \cdot \nabla v \, dx = \int_\Omega f v \, dx.$$

Define

$$a(u, v) := \int_\Omega (\nabla v)^T A(x) \nabla u \, dx \tag{11.2}$$

and note that the right-hand side is nothing but the L^2 inner product

$$\int_\Omega fv\,dx = (f,v)_{0,\Omega}.$$

We note a few remarks:

- $a(u,v)$ is a symmetric, bilinear form
- for given $f \in L^2$, $(f,v)_{0,\Omega} =: l(v)$ is a linear functional
- The weak formulation makes sense for $u, v \in H_0^1$, i.e. those functions $u \in H^1$ whose trace on $\partial\Omega$ vanishes

we define the weak solution for Model Problem 4 as the function $u \in H_0^1(\Omega)$ that satisfies

$$a(u,v) = (f,v)_{0,\Omega}, \qquad \forall v \in H_0^1(\Omega) \tag{11.3}$$

with $a : H^1 \times H^1 \to \mathbb{R}$ defined as in (11.2).

Our task in the next section will be to establish a general theory of well-posedness for equations of the type (11.3). We first show that we have a unique solution to such problems under certain conditions on the bilinear form, which in turn establish a connection to the ellipticity of the underlying partial differential equation. Later we shall extend our analysis to more general problems.

11.2 The General Case

We can take the statement (11.3) as a blueprint for a more general problem formulation, which we shall use to investigate well-posedness. Let H be a Hilbert space, and let $a : H \times H \to \mathbb{R}$ be a bilinear form. For $l : H \to \mathbb{R}$ a linear functional we consider the problem of finding $u \in H$ such that

$$a(u,v) = l(v), \qquad \forall v \in H. \tag{11.4}$$

Note in particular that while the bilinear form (11.2) is symmetric, we do not require this for the general case. We shall show that a few simple conditions suffice for the problem to be well-posed. However, we first need a somewhat abstract, but very useful result.

Theorem 11.1 (Riesz Representation Theorem) *Let H be a Hilbert space, and let $l : H \to \mathbb{R}$ be a continuous linear functional. Then there is a unique $u \in H$, such that*

$$(u,v) = l(v), \qquad \forall v \in H. \tag{11.5}$$

Furthermore, there holds

$$\|l\|_{H^*} = \|u\|_H\,. \tag{11.6}$$

In eq. (11.6) the norm of the functional is defined in the usual way,

$$\|l\|_{H^*} = \sup_{\substack{v\in H\\ v\neq 0}} \frac{|l(v)|}{\|v\|_H}.$$

Proof We first show that there is an element $u \in H$ which minimizes the quadratic functional

$$F(v) := (v,v) - 2l(v).$$

We have

$$\begin{aligned} F(v) &\geq (v,v) - 2C\,\|v\|_H && \text{(Continuity : } |l(v)| \leq C\,\|v\|_H) \\ &\geq -C^2 && \text{(Young's inequality : } ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2) \end{aligned}$$

So F is bounded from below, and we can set $-\infty < d = \inf_{v\in H} F(v)$. To show that a minimum is achieved, consider a sequence $F(v_n) \to d$. By the parallelogram law we have

$$\begin{aligned} \|v_k - v_l\|_H^2 &= 2\,\|v_k\|_H^2 + 2\,\|v_l\|_H^2 - \|v_k + v_l\|_H^2 \\ &= 2\,\|v_k\|_H^2 - 4l(v_k) + 2\,\|v_l\|_H^2 - 4l(v_l) - 4\left\|\frac{v_k+v_l}{2}\right\|_H^2 + 8l\left(\frac{v_k+v_l}{2}\right) \\ &= 2F(v_k) + 2F(v_l) - 4F\left(\frac{v_k+v_l}{2}\right) \\ &\leq 2F(v_k) + 2F(v_l) - 4d \;\to\; 0 \end{aligned}$$

So $(v_k)_{k\in\mathbb{N}}$ is a Cauchy sequence, and as H is complete it must converge to a function $u \in H$. By continuity, we have $F(u) = d$, and the minimum is achieved at u. But this implies that for arbitrary $v \in H$ the function defined by

$$\delta(t) := F(u + tv)$$

has a minimum at $t = 0$, and consequently $\delta'(0) = 0$. Carrying out the differentiation we see that this implies

$$0 = 2(u,v) - 2l(v), \qquad \forall v \in H$$

and this is (11.5).

Now we show uniqueness. Let u_0 and u_1 be such that for all $v \in H$, there holds $(u_0, v) = l(v)$ and $(u_1, v) = l(v)$. Then, taking $v = u_0 - u_1$, there holds

$$\begin{aligned} 0 &= l(u_0 - u_1) - l(u_0 - u_1) \\ &= (u_0, u_0 - u_1) - (u_1, u_0 - u_1) \\ &= \|u_0 - u_1\|_H^2 \end{aligned}$$

and hence $u_0 = u_1$ almost everywhere.

Finally, we show (11.6). From the definition of the operator norm, we have

$$\|l\|_{H^*} = \sup_{0\neq v\in H} \frac{|l(v)|}{\|v\|_H} = \sup_{0\neq v\in H} \frac{|(u,v)|}{\|v\|_H} \leq \|u\|_H\,,$$

where we have used the Cauchy-Schwarz inequality in the last step. We have proved (11.6), if we can show that the reverse inequality also holds. Indeed, we have

$$\|u\|_H^2 = (u,u) = l(u) \leq \|l\|_{H^*}\,\|u\|_H\,.$$

■

With this proof, we can finally show the well-posedness of the weak form of Model Problem 4. We first note the following definitions.

Definition 11.1 *Let $a : H \times H \to \mathbb{R}$ be a bilinear functional on a normed linear space H. We say $a(\cdot,\cdot)$ is bounded (or continuous) if there exists a constant C such that*

$$|a(u,v)| \leq C\,\|u\|_H\,\|v\|_H\,, \qquad \forall u, v \in H \tag{11.7}$$

Definition 11.2 *Let $a : H \times H \to \mathbb{R}$ be a bilinear functional on a normed linear space H. We say $a(\cdot,\cdot)$ is* coercive *on H if there exists a constant $\alpha > 0$, such that*

$$a(v,v) \geq \alpha \|v\|^2, \qquad \forall v \in H. \tag{11.8}$$

Now let H be a Hilbert space and let $a : H \times H \to \mathbb{R}$ be a continuous and coercive bilinear form (not necessarily symmetric). Consider the problem of finding $u \in H$, such that

$$a(u,v) = l(v), \qquad \forall v \in H \tag{11.9}$$

for some continuous linear functional $l : H \to \mathbb{R}$. In this section we show that this problem has a unique solution.

First, by the Riesz representation theorem there is a unique $w_l \in H$, such that

$$l(v) = (w_l, v), \qquad \forall v \in H \tag{11.10}$$

Furthermore, for any *fixed* $u \in H$, we may consider $a(u,\cdot) : H \to \mathbb{R}$ with $v \mapsto a(u,v)$ a linear functional. Again by the Riesz representation theorem there is a unique $w_u \in H$ such that

$$a(u,v) = (w_u, v), \qquad \forall v \in H. \tag{11.11}$$

The subscript u indicates that w depends on u. In fact we can write $w_u = \tau(u)$, where :

Theorem 11.2 *The mapping $\tau : H \to H$, defined by* (11.11) *is a linear operator. Furthermore*

$$\|\tau(v)\| \leq C \|v\|, \tag{11.12}$$

i.e. τ is continuous.

Proof Linearity can be easily seen by direct computation. Indeed, let $\beta, \gamma \in \mathbb{R}$, and $u_1, u_2 \in H$. Then we have

$$\begin{aligned}
(\tau(\beta u_1 + \gamma u_2), v) &= a(\beta u_1 + \gamma u_2, v) \\
&= \beta a(u_1, v) + \gamma a(u_2, v) \\
&= \beta(\tau(u_1), v) + \gamma(\tau(u_2), v) \\
&= (\beta\tau(u_1) + \gamma\tau(u_2), v).
\end{aligned}$$

where the linearity of the bilinear form a, and the linearity of the scalar product have been used.

Now, if $v = 0$, then (11.12) holds trivially. So assume $v \neq 0$, and consider

$$\|\tau(v)\|_H^2 = (\tau(v), \tau(v)) = a(v, \tau(v)) \leq C \|v\| \|\tau(v)\|$$

Dividing by $\|v\|$, yields (11.12). ■

We are interested in the following problem: Is there a unique $u \in H$ such that, for given w_l defined by (11.10), one has

$$(\tau(u), v) = (w_l, v), \qquad \forall v \in H. \tag{11.13}$$

The problem formulated in this manner reads: Is there is a unique u for which the Riesz representer defined in (11.11) is the same as that of the linear functional l. By eq. (11.10) and (11.11)

it is clear that this problem is equivalent to (11.9) The reason we state the problem in this manner, is so that we can apply the Banach fixed-point theorem. Before we show how this is done, we recall:

Theorem 11.3 (Banach fixed-point theorem) *Let $(V, \|\cdot\|)$ be a Banach space, and let $T : V \to V$ be an operator, satisfying for all $v, w \in V$*

$$\|T(w) - T(v)\| \leq L \|w - v\| \tag{11.14}$$

where $0 \leq L < 1$ is a fixed constant. Then there exists a unique fixed point*, i.e. an element $u \in V$, such that*

$$T(u) = u$$

For arbitrary $u_0 \in V$, the iteration

$$u_k = T(u_{k-1}) \tag{11.15}$$

converges to the unique fixed point.

Proof First we show uniqueness. Let u and v both be fixed points of the operator T. Then

$$\|u - v\| = \|T(u) - T(v)\| \leq L \|u - v\|$$

This can only be true if $u = v$, and so any fixed point must be unique.

Now for arbitrary $u_0 \in V$, we get from (11.15),

$$\|u_{k+1} - u_k\| = \|T(u_k) - T(u_{k-1})\| \leq L \|u_k - u_{k-1}\| \leq \cdots \leq L^k \|u_1 - u_0\| .$$

Using this together with the triangle inequality we have for arbitrary $l \in \mathbb{N}$

$$\|u_{k+l} - u_k\| \leq \sum_{i=0}^{l-1} \|u_{k+i+1} - u_{k+i}\| \leq \sum_{i=0}^{l-1} L^{k+i} \|u_1 - u_0\| \leq L^k \sum_{i=0}^{\infty} L^i \|u_1 - u_0\|$$

Using the geometric series, we obtain

$$\|u_{k+l} - u_k\| \leq \frac{L^k}{1 - L} \|u_1 - u_0\| , \qquad \forall l \in \mathbb{N}.$$

The sequence produced by the iteration (11.15) is thus seen to be a *Cauchy sequence*. Since V is complete, it must converge to some $u \in V$. The contraction property (11.14) also implies continuity. So we can pass to the limit in (11.15), and conclude $T(u) = u$. ∎

We are now in a position to prove the main theorem of this chapter:

Theorem 11.4 (Lax-Milgram) *Let H be a Hilbert space. and let $a : H \times H \to \mathbb{R}$ be a continuous and coercive bilinear form, i.e. there holds for some $C > 0$*

$$|a(u, v)| \leq C \|u\|_H \|v\|_H , \qquad \forall u, v \in H, \tag{11.16}$$

and for some $\alpha > 0$

$$a(v, v) \geq \alpha \|v\|_H^2 , \qquad \forall v \in H. \tag{11.17}$$

Let $l : H \to \mathbb{R}$ be a continuous linear functional. Then there exists a unique $u \in H$ solving

$$a(u, v) = l(v), \quad \forall v \in H. \tag{11.18}$$

PROOF Let $\rho = \alpha/C^2$, and define the linear mapping

$$T(v) = v - \rho(\tau(v) - w_l),$$

where $\tau(v)$ and w_l are defined via the Riesz representation (11.11) and (11.10). We show that there is a unique fixed point $T(u) = u$, i.e.

$$u = u - \rho(\tau(u) - w_l) \tag{11.19}$$

which implies $\tau(u) = w_l$ and hence (11.9). Setting $v := v_1 - v_2$ we have

$$\begin{aligned} \|T(v_1) - T(v_2)\|_H^2 &= \|v_1 - \rho(\tau(v_1) - w_l) - v_2 - \rho(\tau(v_2) - w_l)\|_H^2 \\ &= \|v - \rho\tau(v)\|_H^2 \\ &= (v, v) - 2\rho(v, \tau(v)) + \rho^2(\tau(v), \tau(v)) \end{aligned}$$

Noting continuity (11.12), and using coercivity,

$$(v, \tau(v)) = a(v, v) \geq \alpha \|v\|_H^2$$

We can write

$$\begin{aligned} \|T(v_1) - T(v_2)\|_H^2 &\leq \|v\|_H^2 - 2\rho\,\alpha \|v\|_H^2 + C^2\rho^2 \|v\|_H^2 \\ &\leq \left(1 - 2\rho\,\alpha + C^2\rho^2\right) \|v\|_H^2 \\ &= \underbrace{\left(1 - \frac{\alpha^2}{C^2}\right)}_{=:L\in[0,1)} \|v_1 - v_2\|_H^2 \end{aligned}$$

In the last line we have used $0 < \alpha/C \leq 1$ to conclude that $L \in [0, 1)$. (The upper bound is implied by the definition of continuity and coercivity which can only simultaneously be satisfied if $\alpha/C \leq 1$.) By the Banach fixed point theorem, we have a unique fixed point, and the theorem is proved. ■

One should note the generality of the result we have just proved. Any problem that we can formulate in a variational setting, and which leads to a bilinear form being equated to a linear functional, can potentially be shown to possess a unique solution by using the Lax-Milgram Theorem. One merely needs to check whether the assumptions of the theorem are verified. For example, we now show that, in fact, the weak formulation of Model Problem 4 possesses a unique solution.

Example 11.1 We now apply these results to the weak formulation of Model Problem 4, which states that $u \in H_0^1(\Omega)$ is the solution of

$$a(u, v) = l(v), \qquad \forall v \in H_0^1(\Omega),$$

where

$$a(u, v) := \int_\Omega (\nabla v)^T A(x) \nabla u \, dx,$$

and

$$l(v) = \int_\Omega f v \, dx.$$

To apply Theorem (11.4), we need to show that the linear functional and the bilinear form actually satisfy the assumptions of the theorem. The relevant Hilbert space is $H \equiv H_0^1(\Omega)$,

although the restriction to such functions that vanish on the boundary will be relevant only for the coercivity of the bilinear form. Let us show continuity first, and start with the linear functional. Consider for arbitrary $v \in H^1(\Omega)$,

$$|l(v)| \leq \int_\Omega |fv|\,dx \leq \left(\int_\Omega f^2\,dx\right)^{\frac{1}{2}} \left(\int_\Omega v^2\,dx\right)^{\frac{1}{2}} = \|f\|_{0,\Omega}\,\|v\|_{0,\Omega} \leq C\,\|v\|_{1,\Omega}\,,$$

where we have used the Cauchy-Schwarz inequality, and that, by assumption, $f \in L^2(\Omega)$, and thus $\|f\|_{0,\Omega} \leq C$ for some constant $C > 0$. Furthermore, the trivial inequality $\|v\|_{0,\Omega} \leq \|v\|_{1,\Omega}$ has been used.

We now show

$$|a(u,v)| \leq C\,\|u\|_{1,\Omega}\,\|v\|_{1,\Omega}\,, \qquad \forall u,v \in H^1(\Omega).$$

Recall that the functions a_{ij} making up the matrix $A = (a_{ij})_{i,j=1,\ldots,d}$ are bounded. Define $c = \max_{1\leq i,j\leq d} \sup_{x\in\Omega} |a_{ij}(x)|$, and consider, for arbitrary $u,v \in H^1(\Omega)$,

$$\begin{aligned}
|a(u,v)| \leq \int_\Omega \left|(\nabla v)^T A(x)\nabla u\right|\,dx \quad &\leq \quad c\sum_{i,j=1}^d \left|\int_\Omega v_{x_i} u_{x_j}\,dx\right| \\
&\leq \quad c\sum_{i,j=1}^d \|v_{x_i}\|_{0,\Omega}\,\|u_{x_j}\|_{0,\Omega} \\
&= \quad c\left(\sum_{i=1}^d \|v_{x_i}\|_{0,\Omega}\right)\left(\sum_{j=1}^d \|u_{x_j}\|_{0,\Omega}\right) \\
&\leq \quad \widetilde{c}\left(\sum_{i=1}^d \|v_{x_i}\|_{0,\Omega}^2\right)^{\frac{1}{2}}\left(\sum_{j=1}^d \|u_{x_j}\|_{0,\Omega}^2\right)^{\frac{1}{2}} \\
&= \quad \widetilde{c}|u|_{1,\Omega}|v|_{1,\Omega},
\end{aligned}$$

where we have used the Cauchy-Schwarz inequality both for the L^2 inner product and the Euclidean inner product, i.e.

$$\sum_{i=1}^d \|v_{x_i}\|_{0,\Omega} = \sum_{i=1}^d 1\cdot\|v_{x_i}\|_{0,\Omega} \leq \underbrace{\left(\sum_{i=1}^d 1\right)^{\frac{1}{2}}}_{=\sqrt{d}} \left(\sum_{j=1}^d \|u_{x_j}\|_{0,\Omega}^2\right)^{\frac{1}{2}}.$$

Also recall the definition of the semi-norm

$$|u|_{1,\Omega} = \|\nabla u\|_{0,\Omega} = \left(\sum_{j=1}^d \|u_{x_j}\|_{0,\Omega}^2\right)^{\frac{1}{2}}.$$

Clearly, $|\cdot|_{1,\Omega} \leq \|\cdot\|_{1,\Omega}$, and continuity follows.

It seems we have a problem with coercivity. How can we show $a(v,v) \geq \alpha\,\|v\|_{1,\Omega}^2$, if only the gradients appear in the bilinear form? (Note that the norm of the gradient can be zero, without the function being zero.) Fortunately, there is the Poincaré-Friedrichs inequality, which shows that for $v \in H_0^1(\Omega)$ the norm of the gradient is actually a proper norm, not only a semi-norm! For all $v \in H_0^1(\Omega)$ there exists a constant $c > 0$, such that

$$|v|_{1,\Omega} \geq c\,\|v\|_{0,\Omega}$$

(We omit the proof, since it appears intuitively obvious that if a function is zero at the boundary, and the norm of the gradient is zero, it must be zero almost everywhere.)

Now coercivity becomes easy. Let A be uniformly elliptic, according to Definition 10.1. Then

$$a(v,v) = \int_\Omega (\nabla v)^T A(x) \nabla v \, dx \geq \alpha' \int_\Omega |\nabla v|^2 \, dx = \alpha' |v|_{1,\Omega}^2$$

Using the Poincaré-Friedrichs inequality we have for all $v \in H_0^1(\Omega)$,

$$\begin{aligned} a(v,v) &\geq \alpha' |v|_{1,\Omega}^2 \\ &= \frac{\alpha'}{2} \left(|v|_{1,\Omega}^2 + |v|_{1,\Omega}^2 \right) \\ &\geq \frac{\alpha'}{2} \left(|v|_{1,\Omega}^2 + c^2 \|v\|_{0,\Omega}^2 \right) \\ &\geq \frac{\alpha'}{2} \min(1, c^2) \|v\|_{1,\Omega}^2 \end{aligned}$$

and hence the bilinear form is coercive on $H_0^1(\Omega) \subset H^1(\Omega)$. In summary, Theorem 11.4 applies, and the weak formulation of Model Problem 4 has a unique solution.

Recall that not only the existence of a unique solution is part of our concept of a well-posed problem, but also the continuous dependence on the data of the problem. If the bilinear form of the abstract problem (11.4) is coercive, we obtain immediately

$$\alpha \|u\|_H^2 \leq a(u,u) = l(u) \leq \|l\|_{H^\star} \|u\|_H ,$$

where

$$\|l\|_{H^\star} = \sup_{\substack{v \in H \\ v \neq 0}} \frac{|l(v)|}{\|v\|}$$

After a simple rearrangement, we can write

$$\|u\|_H \leq \frac{1}{\alpha} \|l\|_{H^\star} .$$

Adding this to the results we have proved thus far, we can conclude that the abstract problem (11.4) is well-posed if it satisfies the assumptions of the Lax-Milgram theorem.

What about the regularity of the solution? (i.e. how many derivatives does the solution have?) We are guaranteed that $u \in H^1$, as long as we satisfy the prerequisites of the Lax-Milgram theorem. However, we should carefully distinguish between the a priori selection of the space in which we formulate the problem, and the actual regularity of the solution (which may be better in some cases). The former selection is basically guided by an attempt to find the most general setup in which the problem makes sense. However, this does not mean that the solution cannot lie in a "better" space. Investigating the regularity of the elliptic problem, however, is a bit tricky. Note, for example, that up to this point, we have made virtually no assumptions regarding the domain Ω in which we solve the problem. If we wish to know how smooth the solution to problem (11.4) is, we have to specify this more clearly, as well as specify how smooth the coefficient functions are. The results are thus highly problem dependent. We merely cite without proof an important result giving sufficient conditions under which the weak solution of Model problem 4 will at least be in H^2:

Theorem 11.5 *Consider Model problem 4, posed on a* convex *bounded domain* $\Omega \subset \mathbb{R}^d$. *Assume that the coefficient functions are smooth. The weak solution is* H^2 *regular, meaning for every* $f \in L^2(\Omega)$ *there is a solution* $u \in H^2(\Omega)$, *and* $\|u\|_{2,\Omega} \leq C \|f\|_{0,\Omega}$.

11.3 Further Reading

Evans [4] has an alternative proof of the Lax-Milgram theorem in Chapter 6. Also, regularity results are proved there.